

Roberta De Fazio

Curriculum vitae et studiorum

GENERAL INFORMATION	
PERSONAL INFORMATION	Email: roberta.defazio@unicampania.it web: https://www.linkedin.com/in/roberta-de-fazio-1836b2226/
CURRENT POSITION	- July 2026-now : PostDoctoral Researcher at Università della Campania “Luigi Vanvitelli” - September 2026 – now: Mathematics and Physics Professor at Leonardo Da Vinci High School, Aversa, Caserta (Italy) (on leave)
PAST POSITIONS	- Dec 2025 – June 2026: Mathematics and Physics Professor at Russell Newton High School – Scandicci, Florence (Italy) - Dec 2024-Nov 2025: Research Fellow at Università della Campania “Luigi Vanvitelli”. - Dec 2021 – Dec 2024: PhD in Mathematics, Physics and Engineering Applications at the Dept of Mathematics and Physics, Università della Campania “Luigi Vanvitelli” - Sept 2021- Dec 2021: Informatic Consultant at Gartner Italia S.r.l. - July 2019 – August 2019: software developer at SkyIT S.r.l.
PARTICIPATION TO RESEARCH GROUPS	2023-today: member of the research group (SMArt cRiTical SysTems and Software - SMARTISTS) at the Department of Mathematics and Physics of Università della Campania “Luigi Vanvitelli”
OTHER AFFILIATIONS	2025-today: Member of EATCS (European Association for Theoretical Computer Science)
BIBLIOMETRICS	- ORCID https://orcid.org/0000-0002-0271-132X - Google Scholar https://scholar.google.com/citations?hl=it&user=ID_2n6EAAAAJ - Scopus https://www.scopus.com/authid/detail.uri?authorId=57203296376
LANGUAGES	- English fluent speaker and writer - Italian native speaker
EDUCATION	2024-2025 National Habilitation for High School Teaching in Mathematics and Physics (A027) 2021-2024 PhD in Mathematics, Physics and Engineering Applications at the Dept of Mathematics and Physics, Università della Campania “Luigi Vanvitelli”. Dissertation: “Predictive Fault Trees and Data-Driven Techniques in Critical Infrastructures”. 2019-2021 master’s degree in mathematics at Università della Campania “Luigi Vanvitelli” Caserta (Italy). Dissertation: “Neural Networks and Explainability for Protein Tertiary Structure Prediction” – Full mark with honor (110/110 e lode) 2015-2019: Bachelor’s degree in mathematics with Computer Science curriculum at Università della Campania “Luigi Vanvitelli” Caserta (Italy). Dissertation: “Database Project and Development for Morphological Monitoring of the Ground” - Full mark with honor (110/110 e lode)
RESEARCH	
RESEARCH INTERESTS	The research focuses on improving the reliability of critical systems, in particular, focusing on maintenance aspects of large networked transportation infrastructures: - Critical Infrastructures design and modelling (e.g., rail, automotive, avionics, healthcare) - Analysis techniques for critical systems performance and dependability evaluation - Formal models for fault predictions - Causality analysis and Process Mining

	The research activities focus on the design and the implementation of analysis techniques that combine model-based and data-driven approaches in AI-intensive contexts (Clinical Decision Support Systems, Digital Twins, Prediction Maintenance) and on the evaluation and assurance of their trustworthiness. In particular, the efforts of the research are devoted to extending existent formalism like Fault Trees, Petri Nets, and Bayesian Networks and to the integration with Process Mining (traditional and Object-Centric) and machine learning algorithms for Prognostic Health Monitoring, Diagnosis supports and clinical pathway predictions.
NATIONAL AND INTERNATIONAL COLLABORATIONS	1. UHasselt University (Hasselt, BE). The collaboration is framed in an Erasmus internship period in Hasselt from 15th of April 2024 to 15th of July 2024 and is devoted to the study and development of integration tools that enhance the embedding of data-driven techniques, in particular for causality analysis, in model-based framework. 2. University of Chemistry and Technology, Prague. The collaboration is devoted to the study of voice signals and support to the diagnosis of voice disorders using Machine Learning techniques. 3. University of Cassino and Southern Lazio. The collaboration is devoted to the study of Deep Learning technique for image diagnosis, Large Language Models algorithms for medical report analysis and support to clinical decision making. 4. University of Bari "Aldo Moro". The collaboration is devoted to the study of Artificial Intelligence techniques for medical image processing. 5. Gematica S.r.l. (Naples, IT). The collaboration is devoted to the application of analysis techniques on a real case study for a decision support system. The system is developed by the company and applied to the railway networks.
RESEARCH PROJECTS	
PARTICIPATION AND COORDINATION IN LOCAL PROJECTS	1. September 2025 - November 2025: <u>investigator</u> for "MISTICA - sMart lot aneSThesla Cart & cAbinet" (CUP B69J23006090007) from Ministero delle Imprese e del Made in Italy (MIMIT) for the research programme "Accordi di innovazione DM 31/12/2021 secondo bando". 2. 2024 – 2025: <u>WP leader</u> for the WP4 (WP 4: Testing & Validation of the result) for Sistema di sUpporto Decisionale per la diagnosi precoce e il follow-up di pazienti affetti da Valvole Uretrali Posteriori (Decision Support System for the early diagnosis and follow-up of Posterior Urethral Valves) - (CUP B23C24000440001) from IIT/CNR as part of the "RAISE – Robotics and AI for Socio-Economic Empowerment" Ecosystem, which is supported by the Next Generation EU - Italian National Recovery and Resilience Plan (NRRP), Mission 4, Component 2, Investment 1.5, under the call for the creation and enhancement of 'Innovation Ecosystems' and the development of 'Territorial R\&D Leaders' (Directorial Decree No. 2021/3277). 3. 2022-2023: <u>investigator</u> of aDversarial scenArios geneRation With dIgital twiNs In induSTry (DARWNIST), funded by the young researchers programme, Università della Campania "Luigi Vanvitelli", D.R. 834 30-09-2022
PARTICIPATION TO JOURNAL/CONFERENCE SCIENTIFIC COMMITTEE	
TALKS TO CONFERENCE	• "Towards Paediatric Digital Twins: STELLA-Segmentation Tool for Enhanced Localisation and Labelling of Diagnostic Areas" - LDTH Special Session KES2025, September 2025, Osaka (JP) • "Inferring Failure Processes via Causality Analysis: from Event Logs to Predictive Fault Trees" Math4AI&ML- UMI workshop, January 2025, Bari (IT) • "Towards Hepatic Cancer Detection with Bayesian Networks for Patients Digital Twins Modelling" - LDTH Special Session KES2024, September 2024, Sevilla (SP) • "Toward an extension of Fault Trees Formalism in Predictive Maintenance Scenario" – EDCC Conference, April 2024, Leuven (BE) • "CaseID Detection for Process Mining: a heuristic-based methodology" – EDCA workshop, ICPM October 2023, Rome (IT) • "Machine Learning Algorithms comparison: An application on amino acids volume

	- prediction” - PRIMO workshop September 2023, Bari (IT) “Merging Model-Based and Data-Driven Approaches for Resilient Systems Digital Twins Design” - participant talk di ACDL June 2023, Grosseto (IT)
SEMINARS/ INVITED TALKS	May 2nd 2024, seminar: “Predictive Fault Trees and Data-Driven Techniques in Critical Systems”-UHasselt University, Hasselt (BE)- https://www.uhasselt.be/en/onderzoeksgroepen-en/binf/agenda/roberta-de-fazio-predictive-fault-trees-and-data-driven-techniques-in-critical-systems
PROGRAM COMMITTEE MEMBER OF INTERNATIONAL SCIENTIFIC EVENTS	- Special Session Chair of INSIGHT: INtelligent Systems for Imaging-based diaGnosis in HealThcare (IMPACT) - Co-located with the 31st IEEE Symposium on Computers and Communications (ISCC) - June 2026. https://gaetanosettembre.github.io/insight-icts4ehealth-2026/ - Workshop Chair of Innovative Medical image Processing with AI-driven preCision Technologies (IMPACT) -23rd International Conference on Image Analysis and Processing – September 2025 https://impact-workshop.github.io/ - Session Chair of KES Invited Session “2nd edition Leveraging Digital Twins in Healthcare” – September 2025, https://sites.google.com/view/ldth2024/home - Session Chair of KES Invited Session “Leveraging Digital Twins in Healthcare” – September 2024, http://kes2024.kesinternational.org/cms/userfiles/is49.pdf - Member of the technical program committee for IoTBDs Invited Session - Member of the technical program committee for KES Invited Session Intelligent Critical Infrastructures 2023- http://kes2023.kesinternational.org/cms/userfiles/is45.pdf - Member of the organizing committee for the Contemporary Algebraic and Geometric Techniques in Coding Theory and Cryptography virtual summer school, 18-22 July 2022 https://www.matfis.unicampania.it/summer-school
ACTIVITY OF REVIEW	- Discover Artificial Intelligence - RESS: Reliability Engineering and System Safety - BSPC: Biomedical Signal Processing and Control - IM: Intelligence Medicine - Scientific Report - CIBM: Journal of Computers in Biology and Medicine - Discover Applied Sciences - EnPress: Journal of infrastructure, policy and development - Journal of Computational Mathematics and Data Science - MATCOM: Journal of Mathematics and Computers in Simulation
PARTICIPATION TO ERASMUS+ PROGRAMME	April 2024- July 2024: PhD visiting at Hasselt University (Belgium), research collaboration with Professor Benoit Depaire, Business Informatics Research Group
INSTITUTIONAL ACTIVITIES	
INSTITUTIONAL POSITIONS	2021-today: Students’ representative for Direction Committee of the Department of Mathematics and Physics, Università della Campania “Luigi Vanvitelli”.
PARTICIPATION TO ADMISSION/EXAM INATION COMMITTEE OF PHD COURSES	- Jury Member of PhD Dissertation Dr. Greg Van Houdt –UHasselt University, September 2025 - Committe Member of final examination Bachelor degree in Mathematics 19/09/2025 - Committe Member of final examination Bachelor degree in Mathematics 29/01/2025
THIRD MISSION/TECHNOLOGY TRANSFER ACTIVITIES	
	2025: speaker during the Researcher Night for scientific dissemination with SMARTISTS research group. 2019-2025: speaker during Orientation events for higher education institutions – “V:Orienta” at Department of Mathematics and Physics, Università della Campania “Luigi Vanvitelli”. - Expert mentor for the project “V:erso una scelta consapevole: orientare al

	4. riconoscimento delle proprie vocazioni e passioni contrastando false credenze e drop-out” at Istituto Tecnico (ITIS) “Ettore Majorana” in Santa Maria a Vico, Caserta (IT) 2022: speaker during the Researcher Night for scientific dissemination with “PerLaTecnica” association.
SOFTWARE TOOLS¹	1. HECATE - HEpatic CANcer deTectioN via bayesian nEtwork 2. CALYPSO – Causality AnaLYsis for Prediction of System Operation 3. STELLA - Segmentation Tool for Enhanced Localisation and Labelling of Diagnostic Areas
DIDACTIC ACTIVITIES	
TEACHING ACTIVITIES AT UNIV. DELLA CAMPANIA	• 2022-2024: Academic Tutor of “Fondamenti di Informatica” (Fundamentals of Programming) at the bachelor’s degree in “Mathematics” (8 CFU) • 2022-2023: Co-Supervisor Bachelor’s Degree Thesis in Basi di Dati e Sistemi Informativi (Databases and Information Systems) “Improvement of beam quality particles in linear accelerators” – Dott.ssa Erica Ferrone • 2023-2024: Co-Supervisor Bachelor’s Degree Thesis in Basi di Dati e Sistemi Informativi (Databases and Information Systems) “Preprocessing in Clinical Time Series Data: An Application to Anesthesiological Practice” – Dott.ssa Vittoria Di Rosa • 2024-2025: Co-Supervisor Bachelor’s Degree Thesis in Informatica (Computer Science) “Artificial Intelligence Beyond the Black Box: Explainability Techniques for Supporting Clinical Diagnosis” – Dott.ssa Martina Riccardo
NON-INSTITUTIONAL TEACHING	• April/May 2023: lecturer for the “Big Data” course, in collaboration with Consorzio Clara (private funded course).
SCIENTIFIC PUBLICATIONS	
INTERNATIONAL JOURNALS	[J1] R. De Fazio, B. Depaire, S. Marrone, L. Verde - Inferring Failure Processes via Causality Analysis: from Event Logs to Predictive Fault Trees – Journal Reliability and Engineering System Safety (RESS), 2026, DOI: https://doi.org/10.1016/j.ress.2026.112242 [J2] R. De Fazio, S. Marrone, P. Tirelli, R. Chianese, C. Di Nardo, P. Marzuillo, L. Verde- CARE: Clinical AI Predictor for Posterior Urethral Valves - Design, Explainability and Evaluation- Journal of Systems and Software (JSS), 2025, DOI: https://doi.org/10.1016/j.jss.2025.112720 [J3] Vrba, J. Steinbach, T. Jirsa, L. Verde, R. De Fazio, Y. Zeng, K. Ichiji, L. Hájek, Z. Sedláková, Z. Urbániová, M. Chovanec, J. Mareš, N. Homma; Reproducible Machine Learning-Based Voice Pathology Detection: Introducing the Pitch Difference Feature; Journal of Voice, 2025, DOI: https://doi.org/10.1016/j.jvoice.2025.03.028 [J4] L. Verde, F. Marulli, R. De Fazio, L. Campanile, S. Marrone - HEAR set: A ligHtwEight acoustic paRameters set to assess mental health from voice analysis; Computers in Biology and Medicine, vol- 182, 2024, DOI: https://doi.org/10.1016/j.compbimed.2024.109021 [J5] Marrone, S., Campanile, L., De Fazio, R., Di Giovanni, M., Gentile, U., Marulli, F., Verde, L.; A Petri net oriented approach for advanced building energy management systems; Journal of Ambient Intelligence and Smart Environments, 15 (3), pp. 211-233. DOI: 10.3233/AIS-230065 [J6] De Fazio R., Di Giovannantonio R., Bellini E., Marrone S.; Explainability Comparison between Random Forests and Neural Networks—Case Study of Amino Acid Volume Prediction; Information (Switzerland), 14 (1), art. no. 21, DOI: 10.3390/info14010021.

¹ All the software, available at <https://github.com/robidfz>

INTERNATIONAL CONFERENCE PROCEEDINGS	[C1] C. Russo, G. Settembre, G. Gargano, M.S. de Biase, R. De Fazio - AI in Pediatric Urology: Deep Learning-based Approach supporting Posterior Urethral Valves Diagnosis on VCUG Imaging - Image Analysis and Processing - ICIAP 2025 Workshops. ICIAP 2025. Lecture Notes in Computer Science, vol 16170. Springer - DOI: https://doi.org/10.1007/978-3-032-11381-8_12 [C2] R. De Fazio, M.S. de Biase, P. Marzuillo, P. Tirelli, F. Marulli, S. Marrone, L. Verde - Towards Paediatric Digital Twins: STELLA-Segmentation Tool for Enhanced Localisation and Labelling of Diagnostic Areas – Procedia Computer Science, vol. 270, pp.5579-5588 - DOI: 10.1016/j.procs.2025.10.026 [C3] M.S. de Biase, R. De Fazio, S. Marrone -Detecting Vegetation Patterns in Satellite Images: SILVIA, a Segmentation-Based Approach – Procedia Computer Science, vol. 270, pp.5520-5530 - DOI: 10.1016/j.procs.2025.10.020 [C4] S. Marrone, R. De Fazio, R. Picone, L. Verde -Towards a pre-surgery clinical decision support system – Procedia Computer Science vol. 270, pp.5559-5568- DOI: 10.1016/j.procs.2025.10.024 [C5] C. Napolino, R. De Fazio, S. Marrone, L. Verde - Towards a Digital Twin of the Cardiovascular System; Proceedings of the 10th International Conference on Internet of Things, Big Data and Security –2025 DOI: 10.5220/0013520700003944 [C6] R. De Fazio, A. Bartos, V. Leonetti, S. Marrone, L. Verde; Towards Hepatic Cancer Detection with Bayesian Networks for Patients Digital Twins Modelling; Procedia Computer Science, vol.246, pp. 5104-5113, 2024, DOI: https://doi.org/10.1016/j.procs.2024.09.598 [C7] L. Campanile, R. De Fazio, M. Di Giovanni, F. Marulli - Beyond the hype: Toward a concrete adoption of the fair and responsible use of AI – CEUR workshop Proceedings, 3762, pp. 60–65, 2024, [C8] R. De Fazio, A. Balzanella, S. Marrone, F. Marulli, L. Verde, V. Reccia, and P. Valletta, "CaseID detection for process mining: a heuristic-based methodology," Lecture Notes in Business Information Processing, vol.503, 2023 – DOI: https://doi.org/10.1007/978-3-031-56107-8_4 [C9] Lelio Campanile, Roberta De Fazio, Michele Di Giovanni, Stefano Marrone, Fiammetta Marulli, Laura Verde; Inferring Emotional Models from Human-Machine Speech Interactions; Procedia Computer Science, vol. 225, pp. 1241–1250, 2023, DOI: https://doi.org/10.1016/j.procs.2023.10.112 [C10] Lelio Campanile; Maria Stella de Biase; Roberta De Fazio; Michele Di Giovanni; Fiammetta Marulli; Laura Verde; Merging model-based and data-driven approaches for resilient systems digital twins design; 2023 IEEE International Conference on Cyber Security and Resilience (CSR) 2023- DOI: 10.1109/csr57506.2023.10224945
BOOK CHAPTERS	[B1] De Fazio, R., Napolino, C., Marrone, S., Verde, L. (2025). A Unifying Approach for Digital Twins in Healthcare: Perspectives and Experiences. In: Jain, A., Lim, C.P., Jain, L.C. (eds) Digital Transformation in Healthcare Systems for Patient Care. Intelligent Systems Reference Library, vol 279. Springer, Cham. https://doi.org/10.1007/978-3-031-95044-5_8

Attesto sotto la mia responsabilità, ai sensi e per gli effetti dell'articolo 47 del d.p.r. 445/2000 e consapevole di quanto prescritto dagli articoli 75 e 76 del medesimo d.p.r., rispettivamente sulla responsabilità penale prevista per chi rende false dichiarazioni e sulla decadenza dai bene ci eventualmente conseguenti al provvedimento emanato sulla base di dichiarazioni non veritiere, la veridicità delle informazioni contenute nel presente curriculum.

Prato, 12/11/2025

Roberta De Fazio